Dynamic Programming

1.

Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of "axyb" and "abyxb". After hours of hard work, she has made this LCS table, M.

	empty	a	b	y	x	b
empty	0	0	0	0	0	0
a	0	1	1	1	1	1
x	0	1	1	1	2	2
y	0	1	1	2	2	2
b	0	1	2	2	2	3

a. CO1 **Explain** what you understand by the value M[3][4] = 2. **2**

b. CO2 **Find** out the LCS String from the table. Show the steps of your work. **3**

c. CO4 **Determine** the maximum profit for the 0-1 Knapsack problem given in the following table using Dynamic Programming. Show the steps with a recursion tree or the memory matrix. **5**

Knapsack Weight: 8 kg

Objects	***i***	***ii***	***iii***	***iv***
Weight (kg)	5	4	6	3
Profit ($)	11	10	12	9

2.

Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of "axyb" and "axyxb". After hours of hard work, she has made this LCS table, M.

	empty	a	x	y	x	b
empty	0	0	0	0	0	0
a	0	1	1	1	1	1
x	0	1	2	2	2	2
y	0	1	2	3	3	3
b	0	1	2	3	3	4

a. CO1 **Explain** what you understand by the value M[3][4] = 3. **2**

b. CO2 **Find** out the LCS String from the table. Show the steps of your work. **3**

c. CO4 **Determine** the maximum profit for the 0-1 Knapsack problem given in the following table using Dynamic Programming. Show the steps with a recursion tree or the memory matrix. **5**

Knapsack Weight: 8 kg

Objects	***i***	***ii***	***iii***	***iv***
Weight (kg)	5	4	6	3
Profit ($)	9	8	20	9

3.

Suppose you went to Miniso to buy a Teddy Bear Plushie (a soft toy or doll) as a gift for your younger sister's birthday. The plushie is 700 BDT. At Miniso, all items are subject to a 6% additional VAT charge.

You bought one plushie and gave the cashier 750 BDT. The cashier has a huge supply of 1 taka, 2 taka, and 5 taka coins in the cashbox. You don't want to carry many coins, so you asked her to return the change using a minimum number of coins.

CO4	i)	Using the Dynamic Programming approach, **determine** how many coins she should return in this scenario.	**06**
CO4	ii)	Which coins did you get from the cashier? **Show** how your table was used to pick up these coins. You may use arrows and circles to point to the chosen cells.	**02**
CO7	iii)	Suppose you did not apply memoization to find the answer for question (i). What will be the time complexity of that brute force approach? **Explain** with proper reasoning.	**02**

4.

A team of two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the Louvre are too precious, and taking a fraction of any item won't sell in the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Louvre with an empty knapsack weighing a total of 8 kg. Despite the fact that both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Louvre Museum are listed below.

Objects	**Jewelry**	**Sculpture**	**Painting**	**Book**	**Mummy**
Profit ($)	5	9	5	4	6
Weight (Kg)	3	5	4	1	12

CO4	i)	**Calculate** the maximum profit Denver can make using his strategy. What items did he pick up? **Show** how Denver used the DP table to select these objects. You may use arrows and circles to point to the chosen cells.	**08**
CO4	ii)	Does Nairobi's belief remain valid after the robbery? **Prove** it.	**02**

5.

Two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the museum are too precious, and taking a fraction of any item won't sell on the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Royal Treasury with an empty knapsack weighing a total of **7 kg** each. Even though both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Royal Treasury Museum are listed below.

Objects	Diamond	Jewelry	Sculpture	Painting	Gold Crest
Profit ($)	3	4	12	9	12
Weight (kg)	1	2	8	4	5

CO2 a) **Simulate** your dynamic programming algorithm to find the maximum profit Denver can make.

CO2 b) From your memory/dp table, **find** out which items would he take to make this amount of profit?

CO3 c) Does Nairobi's belief remain valid after the robbery? **Prove** it.

6.

You are Monco, a video game character, you have just stepped into level 2. This game requires you to acquire necessity items to sustain through a level and this is done by trading reward points from the previous level. From level 1 you have gained 9,000 points. Below is a list of items you can get by trading your reward points. Each item has certain properties but be wise to pick the ones which will maximize your sustainability by maximizing mission impact.

	Gold Bar	Pain Killer	Waterproof Boots	Sun glasses	Energy Bar	Night Vision
Trade Price (points)	8000	2000	2000	2000	1000	3000
Mission Impact	1	5	3	2	2	4

CO2 a) **Simulate** your chosen algorithm on the below items to find your picked items. **07**

CO3 b) If you were to apply brute force, what would be the time complexity and why? **Explain.** **03**
Assume that points you have gained are denoted as P, and the number of items is N.

7.

Karen started a website that displays different advertisements. She wants to maximize her advertisement revenue. Each month she receives a list of advertisements, each with a certain value (revenue) and a size (space taken on the webpage). The webpage has a limited amount of space, and she selects a subset of advertisements to display in order to maximize her total revenue, while staying within the available space. No advertisement is selected more than once, and each advertisement that is selected has to be displayed in full.

For example, following is a list of advertisements she received for the month of August:

Advertisements: Ad 1 - Value: $10, Size: 2 Ad 2 - Value: $8, Size: 1 Ad 3 - Value: $15, Size: 3 Ad 4 - Value: $6, Size: 2 Maximum Space Available: 5	Solution: Select Advertisements: Ad 1, Ad 3 Total Revenue: $10 + $15 = $25 Space Occupied = 2 +3 = 5

a) [CO1] Suppose, you are trying to come up with a Dynamic Programming approach to solve this problem. You build a table, each cell representing the solution (optimal revenue value) to a subproblem. **Give** the formula for filling up each entry of the table. **02**

b) [CO1] Suppose in the month of September, she received the following list of advertisements: **06**

Advertisements:
Ad 1 - Value: $12, Size: 4
Ad 2 - Value: $8, Size: 2
Ad 3 - Value: $9, Size: 3
Ad 4 - Value: $7, Size: 2
Ad 5 - Value: $5, Size: 1

The maximum available space is 5.

Fill the following table according to the formula you developed in question (a). Then write the set of advertisements Karen should select.

Value	Size	Item	Maximum Available Space					
			0	1	2	3	4	5
-	-	0						
12	4	1						
8	2	2	0	0	8	8	12	12
9	3	3						
7	2	4						
5	1	5						

c) [CO3] Suppose, Karen discovered that the top half of some of the advertisements contain nothing. So, from the month of October she is implementing a new feature, the size of at most K items can be changed to half of its original size. Revenue will stay the same. For example, **02**

There are four advertisements to select from,
Value = [10, 12, 8, 6]
Required Space = [4, 5, 6, 6]
Max Available Space = 7
K = 1

Output: 22
Explanation: Without changing the required space of any item, the solution would be taking only Ad-2, giving a revenue of 12. But if we change the size of Ad-1 into half of its original size, we can take both Ad-1 and 2. Then the total revenue is 22, which is the maximum.

Think about how you would solve this problem using a Dynamic Programming approach. Then, **write** a recursive formulation of your solution or **explain** your idea in pseudocode/flow-chart/step-by-step instructions format.

8.

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

CO1 a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses. **04**

CO1 b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. Note that, selecting a course still means taking the whole, with full credits. **02**

CO1 c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message. **04**

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Courses, credits and values are provided below:

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Value	10	8	16	13	17

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10.

You are given two strings, '**cbabdcb**' and '**badcbc**', and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.

CO3 a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer. **05**

11.

You are given two strings, '**abcdefg**' and '**bfegba**', and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.

a. Find the longest common sequence using the approach proposed by person 'X'. **Show** the necessary **simulation** to find the answer. **05**

12.

You are now attending the CSE221 final exam and there is a limited time to solve the questions. Each question requires a different amount of time and offers varying marks. After answering some questions, you suddenly realized that only 6 minutes remained. You aim to maximize the total marks you can score, given the constraints on time.

Questions and their properties:

- Question 4: Time = 5 minutes, Marks = 8
- Question 5: Time = 3 minutes, Marks = 7
- Question 6: Time = 4 minutes, Marks = 12
- Question 7: Time = 6 minutes, Marks = 10
- Question 8: Time = 2 minutes, Marks = 6
- Question 9: Time = 1 minute, Marks = 5

a. **CO3** If your faculty doesn't give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. **Show** a full detailed calculation. **05**

b. **CO3** Fortunately, your faculty enters the hall and announces a new rule: partial marks will be given for partially completed answers. If you can solve part of a question, you'll get marks proportional to the time you spend on it. For example, solving half of a 4-minute question gives you half the marks. **05**

Now, with this new opportunity, how does your strategy shift? How will you maximize your score within the time limit, factoring in that you can now earn something even if you don't fully complete a question? **Calculate** the maximum marks you can score with this fortunate announcement. Did you score more than the previous scenario?

c. **CO2** How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly. **02**

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Questions and their properties:

- Question 4: Time = 4 minutes, Marks = 7
- Question 5: Time = 2 minutes, Marks = 9
- Question 6: Time = 3 minutes, Marks = 10
- Question 7: Time = 5 minutes, Marks = 8
- Question 8: Time = 6 minutes, Marks = 12
- Question 9: Time = 1 minute, Marks = 3

a. **CO3** If your faculty doesn't give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. **Show** a full detailed calculation. **04**

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c. **CO2** How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly. **02**

14.

Answer the following questions:

a. Can merge sort be written as a dynamic programming algorithm? **Explain** your answer. **02**
CO2

b. We are calculating the longest common subsequence (LCS) between two strings, $S = X_1X_2X_3X_4$ and $T = Y_1Y_2Y_3Y_4$. To do this, we fill an array C where $C_{i,j}$ represents the length of the LCS between the prefix of length i from S and the prefix of length j from T.

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♡
X_4	0	1	♣	♠	◇

CO1
i. What does the cell $C_{0,0}$ represent? **01**
ii. What can be said about the characters X_1 and Y_4? **01**
iii. What can be said about the characters X_2 and Y_4? **01**
iv. What is the value of ♡? **01**
v. What is the value of ♣? **01**
vi. When we are filling up the i-th row of our dynamic programming table C, what rows do we need to have access to? **01**
CO3 vii. Given the observation above can we optimize our space complexity further? **02**

15.

b. We are calculating the longest common subsequence (LCS) between two strings, $S = X_1X_2X_3X_4$ and $T = Y_1Y_2Y_3Y_4$. To do this, we fill an array C where $C_{i,j}$ represents the length of the LCS between the prefix of length i from S and the prefix of length j from T.

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♡
X_4	0	1	♣	♠	◇

CO1
i. What does the cell $C_{0,1}$ represent? **01**
ii. What can be said about the characters X_2 and Y_3? **01**
iii. What can be said about the characters X_3 and Y_3? **01**
iv. What is the value of ♣? **01**
v. What is the value of ♠? **01**
vi. When we are filling up the i-th column of our dynamic programming table C, what columns do we need to have access to? **01**
CO3 vii. Given the observation above can we optimize our space complexity further? **02**

16.

DNAs contain genetic and hereditary information of almost every living organism. DNA Sequence Matching is a procedure to compare two DNA sequences and find similarity between them. Given below is the recurrence relation of a dynamic programming algorithm to perform sequence matching of two DNAs.

$$seq(i,j) = \max \begin{cases} seq(i, j-1) - 1 \ , \\ seq(i-1, j) + p(i, j) \ , \\ seq(i-1, j) - 1 \end{cases}$$

$where,$

$$seq(i,j) = 0, \quad if\ i = 0\ or\ j = 0$$
$$p(i,j) = 1, \quad if\ DNA1[i] = DNA2[j]$$
$$p(i,j) = -2, \quad otherwise$$

As you can see, the formula takes two indices, $i\ and\ j$ of two DNA sequences given as input. Then it produces a similarity score as output. We want to apply this formula to compare the following DNA sequences.

index	1	2	3	4	5
DNA1	G	C	G	T	A
DNA2	C	T	G	A	G

Now answer the following questions:

a) [CO1] Put appropriate values to **fill the gaps** in the line below (answer in your script): **02**
$seq(__, __)\ represents\ the\ final\ similarity\ score\ of\ the\ given\ DNAs.$

b) [CO1] **Apply** the formula to calculate a similarity score between the DNA sequences given above. Show your work with either a recursion tree or a memory table. **05**

c) [CO2] **State** the time complexity of this algorithm with proper reasoning. **02**
You can assume that, the lengths of the DNA sequences are $N\ and\ M$ respectively.

d) [CO3] Mriaslun, a talented algorithmist, claims that he can implement the algorithm in an optimized way so that it only takes $O(min(N, M))$ space. Can you do it too? **Explain** your idea. **01**

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$$p(i,j) = -2, \quad otherwise$$

As you can see, the formula takes two indices, $i\ and\ j$ of two DNA sequences given as input. Then it produces a similarity score as output. We want to apply this formula to compare the following DNA sequences.

index	1	2	3	4	5	6
DNA1	G	A	T	C		
DNA2	C	T	G	A	G	C

Now answer the following questions:

a) [CO1] Put appropriate values to **fill the gaps** in the line below (answer in your script): **02**
$seq(__, __)$ *represents the final similarity score of the given DNAs.*

b) [CO1] **Apply** the formula to calculate a similarity score between the DNA sequences given above. Show your work with either a recursion tree or a memory table. **05**

c) [CO2] **State** the time complexity of this algorithm with proper reasoning. **02**
You can assume that, the lengths of the DNA sequences are $N\ and\ M$ respectively.

d) [CO3] Mriaslun, a talented algorithmist, claims that he can implement the algorithm in an optimized way so that it only takes $O(min(N, M))$ space. Can you do it too? **Explain** your idea. **01**

18.

a. String1: <Your ID>
CO1 String2: _____

i. Create String2 in a way that the following constraints match. **Write** the string. **03**
   - Length of String2 = 8 (eight)
   - Length of LCS(String1,String2) = 5 (five)
   - There are exactly three distinct LCS (of length = 5)

ii. **Write** all three LCS strings (just write, no need to show any work). **01**

b. String1: <Your ID>
CO1 String2: "20251905"

iii. **Find** the length of the LCS of these two strings. You can use a memory table or a recursion tree. Show your work. **04**

iv. Using your work from the previous question, **find** the LCS string. **02**